Chapter 24: Measuring a Nation's Income

Learning Objectives

- 24.1 Calculate how an economy's total income equals its total expenditure.
- 24.2 Define and calculate gross domestic product (GDP).
- 24.3 Identify and break down GDP into its four major components.
- 24.4 Define and calculate real GDP and nominal GDP.
- 24.5 Evaluate whether GDP is a good measure of economic well-being.

Key Points

- Because every transaction has a buyer and a seller, the economy's total expenditure must equal its income.
- Gross domestic product (GDP) measures an economy's total expenditure on newly produced goods and services and the total income earned from the production of these goods and services. More precisely, GDP is the market value of all final goods and services produced within a country in a given period.
- GDP is divided among four components of expenditure: consumption, investment, government purchases, and net exports. Consumption includes spending on goods and services by households, with the exception of purchases of new housing. Investment includes spending on business capital, residential capital, and inventories. Government purchases include spending on goods and services by local, state, and federal governments. Net exports equal the value of goods and services produced domestically and sold abroad (exports) minus the value of goods and services produced abroad and sold domestically (imports).
- Nominal GDP uses current prices to value the economy's production of goods and services. Real GDP uses constant base-year prices to value this production. The GDP deflator—calculated from the ratio of nominal to real GDP—measures the level of prices in the economy.
- GDP is a good measure of economic well-being because people usually prefer higher incomes to lower incomes. But it is not a perfect measure of well-being. For example, GDP excludes the value of leisure and the value of a clean environment.

Income and Expenditure

- **Gross Domestic Product (GDP)** measures total income of everyone in the economy.
- GDP also measures total expenditure on the economy's output of goods and services.

For the economy as a whole,
income equals expenditure
because every dollar a buyer spends
is a dollar of income for the seller.

Gross Domestic Product (GDP) Is…

…the market value of all final goods & services produced within a country in a given period of time.

- Goods are valued at their market prices, so:
  - All goods measured in the same units (e.g., dollars in the U.S.)
  - Things that don't have a market value are excluded, e.g., housework you do for yourself.

Gross Domestic Product (GDP) Is…

…the market value of all final goods & services produced within a country in a given period of time.

- **Final goods**: intended for the end user
- **Intermediate goods**: used as components or ingredients in the production of other goods
- GDP only includes final goods—they already embody the value of the intermediate goods used in their production.

Gross Domestic Product (GDP) Is…

…the market value of all final goods & services produced within a country in a given period of time.

- GDP includes tangible goods (like DVDs, mountain bikes, beer)
- and intangible services (dry cleaning, concerts, cell phone service).

Gross Domestic Product (GDP) Is…

…the market value of all final goods & services produced within a country in a given period of time.

- GDP includes currently produced goods, not goods produced in the past.

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Gross Domestic Product (GDP) Is…

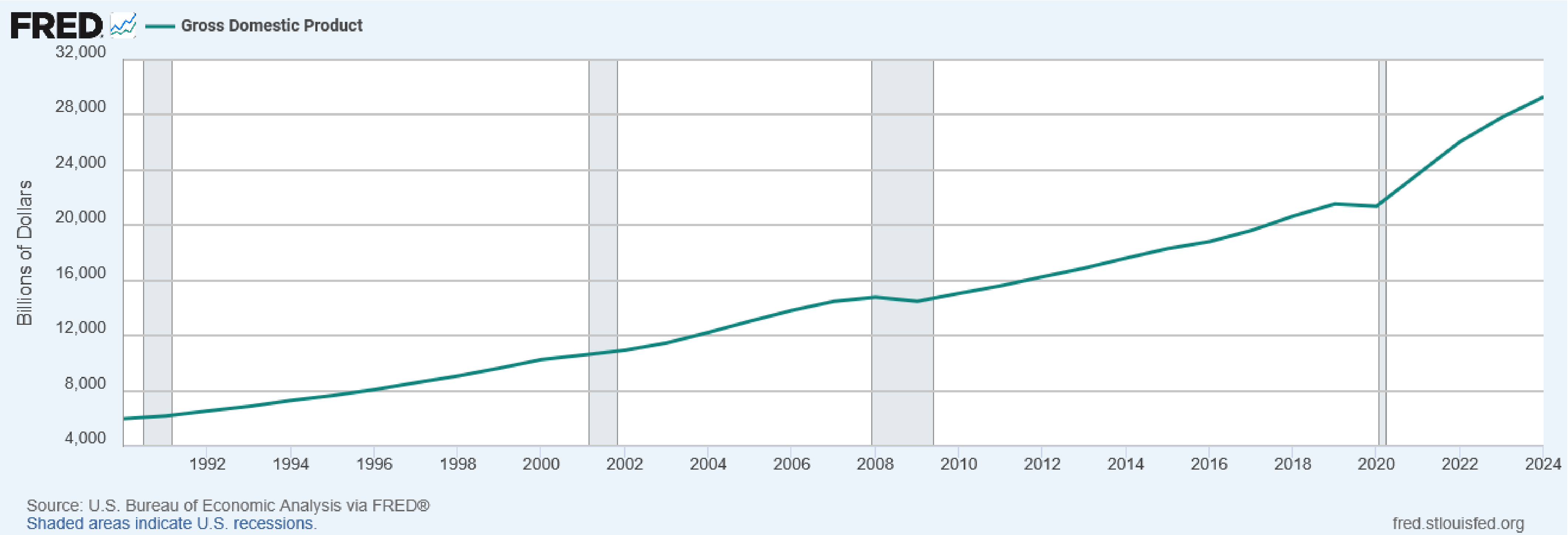
…the market value of all final goods & services produced within a country in a given period of time.

- GDP measures the value of production that occurs within a country's borders, whether done by its own citizens or by foreigners located there.
- Gross National Product would include income earned by citizens abroad.

Gross Domestic Product (GDP) Is…

…the market value of all final goods & services produced within a country in a given period of time.

Usually a year or a quarter (3 months)



Activity: Does it Count?

- Classify each scenario as **Counted in GDP** or **Not Counted in GDP**, and identify *why* (e.g., final vs. intermediate good, market vs. non-market activity, used goods, etc.).

Scenario

1. You buy a new laptop made in the U.S.
2. You sell your old phone on Facebook Marketplace.
3. A farmer eats vegetables from her own garden.
4. A government hires workers to repair roads.
5. A U.S. firm builds a factory in Mexico.
6. A foreign tourist stays in a New York hotel.

Activity: Does it Count?

Scenario	Counted in GDP?	Reason
1. You buy a new laptop made in the U.S.	Yes	Final good, newly produced in domestic economy.
2. You sell your old phone on Facebook Marketplace.	No	Used good; already counted in earlier GDP.
3. A farmer eats vegetables from her own garden.	No	Non-market production, not part of market transactions.
4. A government hires workers to repair roads.	Yes	Government spending on goods and services.
5. A U.S. firm builds a factory in Mexico.	No	Investment abroad, not domestic production.
6. A foreign tourist stays in a New York hotel.	Yes	Export of U.S. services.

The Components of GDP

- Recall: GDP is total spending.
- Four components:
  - Consumption (**C**)
  - Investment (**I**)
  - Government Purchases (**G**)
  - Net Exports (**NX**)
- These components add up to GDP (denoted **Y**):

$$\mathbf{Y = C + I + G + NX}$$

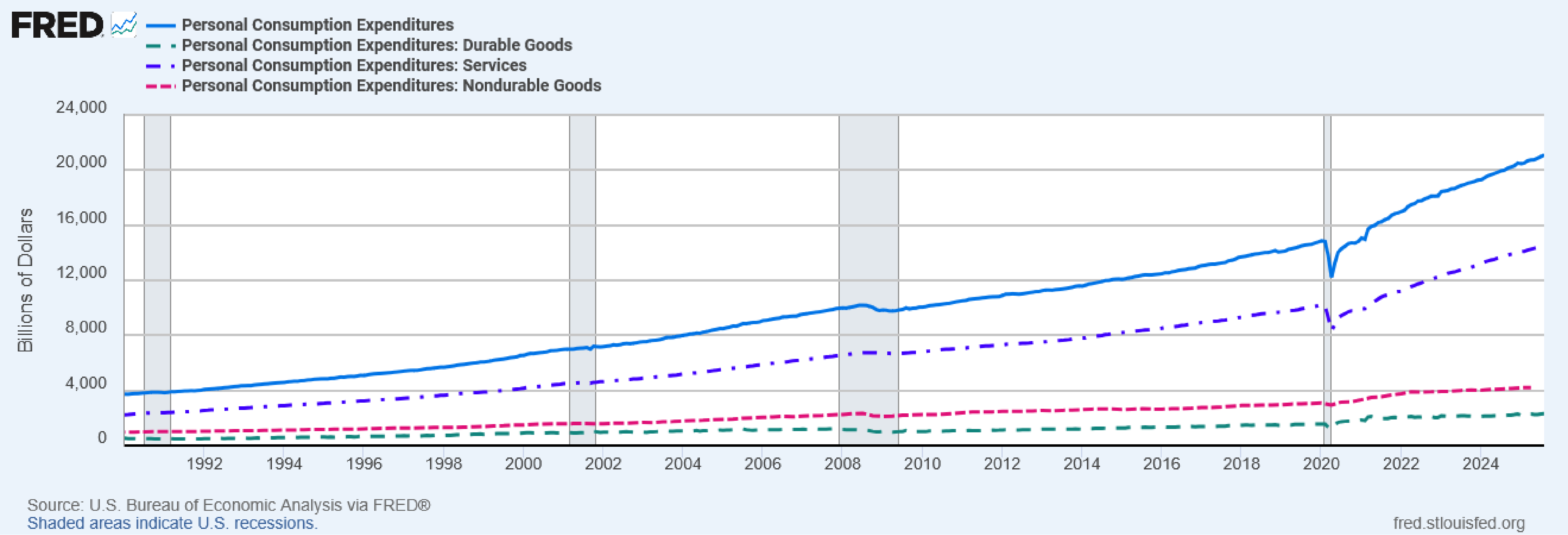
Consumption (C)

- Is total spending by households on goods and services.
- Note on housing costs:
  - For renters,
    consumption includes rent payments.
  - For homeowners,
    consumption includes the imputed rental value of the house, but not the purchase price or mortgage payments.

Consumption (C)

- Individual components of consumption include
  - Durable goods (10%)
    - Goods expected to last longer than 3 years
      - cars, TVs, refrigerators
  - Nondurable Goods (20%)
    - Goods with a life expectancy of less than 3 years
      - food, clothing
  - Service (70%)
    - haircuts, attorney fees, dentist visits

Consumption (C)



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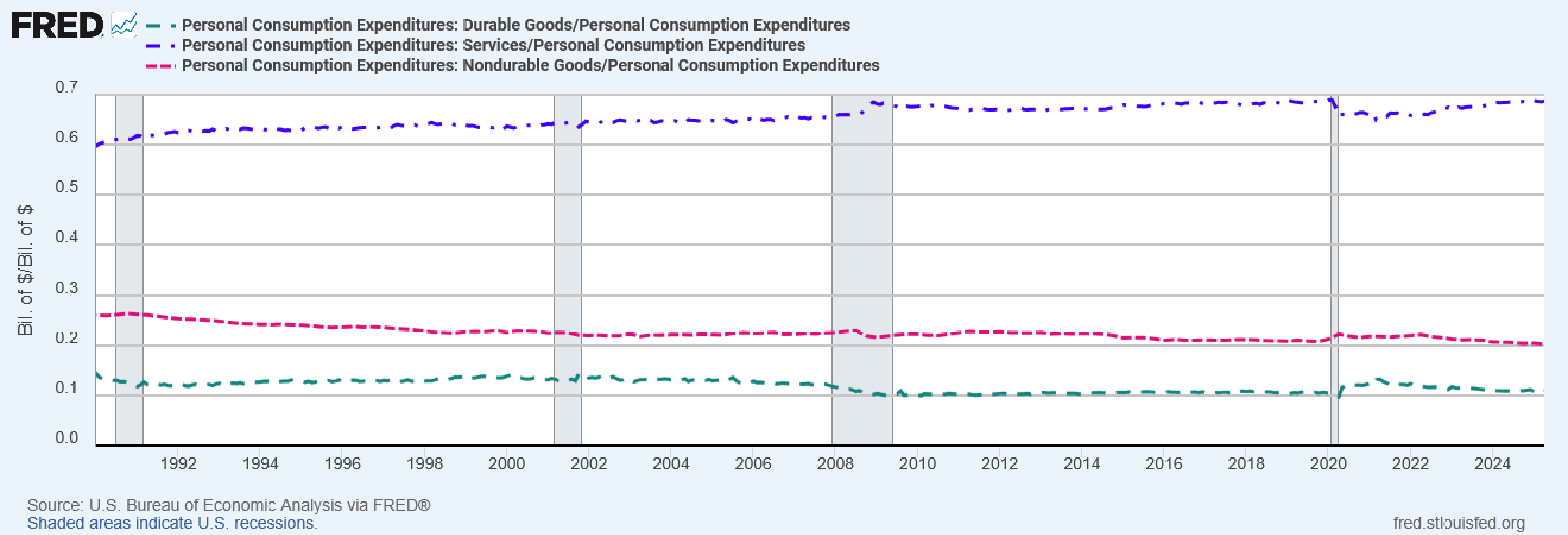
Consumption (C)



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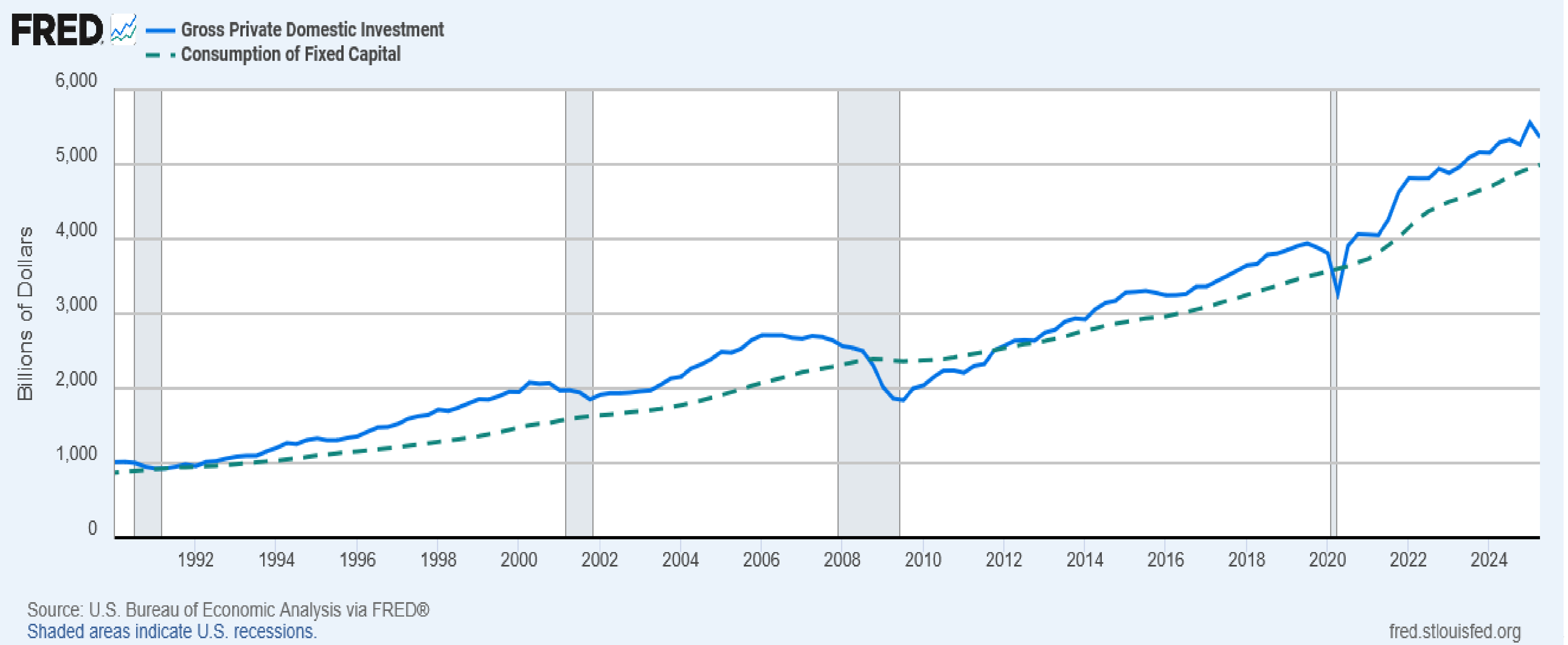
Investment (I)

- is total spending on goods that will be used in the future to produce more goods.
- includes spending on
  - **capital equipment** (e.g., machines, tools)
  - **structures** (factories, office buildings, houses)
  - **inventories** (goods produced but not yet sold)

*Note: **"Investment"** does not mean the purchase of financial assets like stocks and bonds.*

Investment (I)

- Investment also includes depreciation (consumption of fixed capital)
  - Taking depreciation away from investment gives net investment
  - As long as I > D the economy's capital stock is growing



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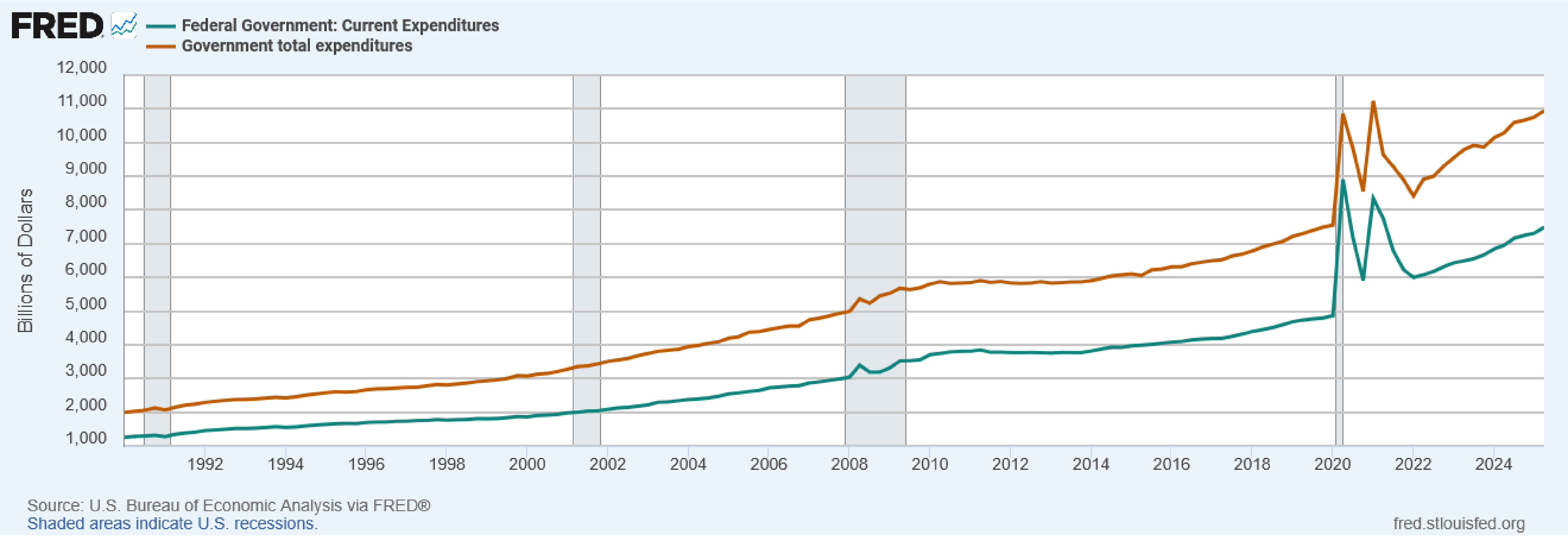
Government Purchases (G)

- is all spending on the goods & services purchased by government at the federal, state, and local levels.
- **G** excludes **transfer payments**, such as Social Security or unemployment insurance benefits.
    - They are not purchases of goods & services.

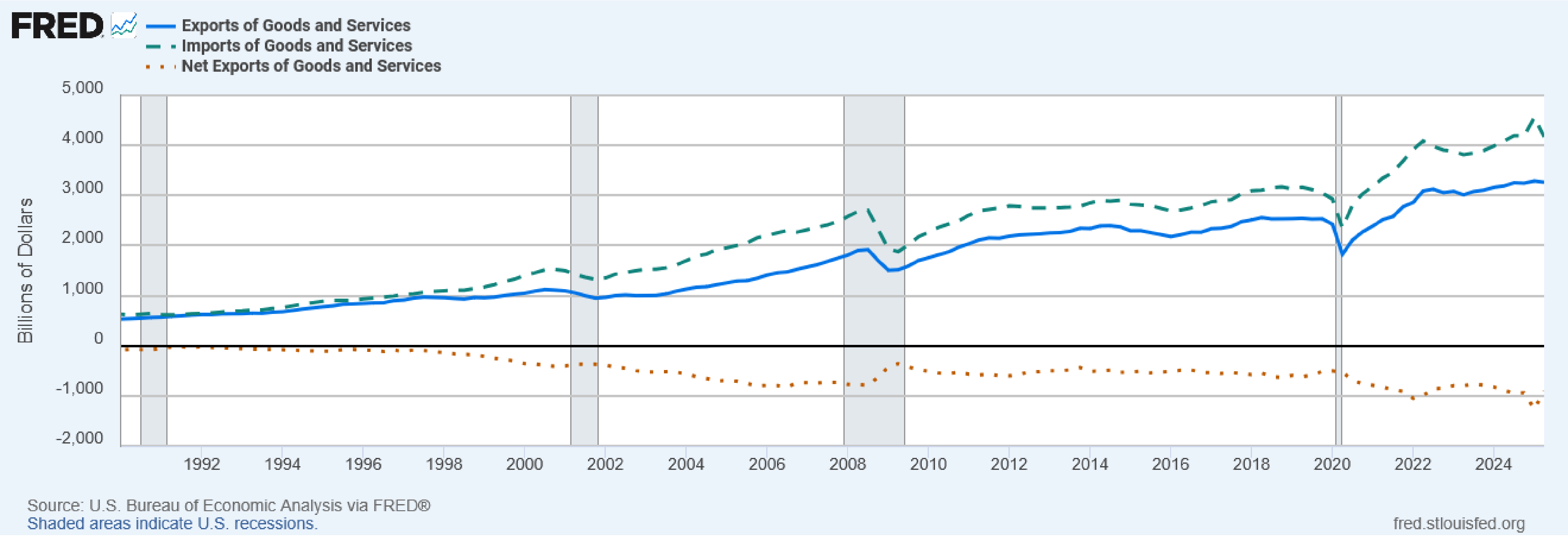
Government Purchases (G)



Net Exports (NX)

- **NX** = exports – imports
- Exports represent foreign spending on the economy's goods & services.
- Imports are the portions of **C**, **I**, and **G** that are spent on goods & services produced abroad.

Net Exports (NX)



GDP and Its Components (2024)

Y = C + I + G + NX

	billions	*% of GDP*	*per capita*
Y	$29,298	100.0	$86145.25
C	19,896	67.9	58,500.44
I	5,259	17.9	15,463.1
G	10,424	35.5	30,649.81
NX	–898.4	–3.1	–2,641.58

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Activity: GDP and Its Components

Household purchases of durable goods	$1293
Household purchases of nondurable goods	$1717
Household purchases of services	$301
Household purchases of new housing	$704
Purchases of capital equipment	$310
Inventory changes	$374
Purchases of new structures	$611
Depreciation	$117
Salaries of government workers	$1422
Government expenditures on public works	$553
Transfer payments	$777
Foreign purchases of domestically produced goods	$88
Domestic purchases of foreign goods	$120

1. What is consumption (C)?
2. What is investment (I)?
3. What are government expenditures (G)?
4. What are net exports (NX)?

Activity: GDP and Its Components

1. Consumption = Durable Goods + Nondurable Goods + Services
   = $1293 + $1717 + $301
   = $3331

2. Investment = New Housing + Capital + Inventory + New Structures
   = $704 + $310 + $374 + $611
   = $1999

3. Government Exp. = Salaries + Expenditures
   = $1422 + $553
   = $1975

4. Net Exports = Foreign Purchases – Domestic Purchases
   (Exports) (Imports)
   = $88 - $120
   = - $32

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Activity: How is GDP Affected?

In each of the following cases, determine how much GDP and each of its components is affected (if at all).

A. Debbie spends $200 to buy her husband dinner at the finest restaurant in Boston.

B. Sarah spends $1800 on a new laptop to use in her publishing business. The laptop was built in China.

C. Jane spends $1200 on a computer to use in her editing business. She got last year's model on sale for a great price from a local manufacturer.

D. General Motors builds $500 million worth of cars, but consumers only buy $470 million worth of them.

Activity: How is GDP Affected?

A. Consumption and GDP rise by $200.

B. Investment rises by $1800, net exports fall by $1800, GDP is unchanged.

C. Current GDP and investment do not change, because the computer was built last year.

D. Consumption rises by $470 million, inventory investment rises by $30 million,
and GDP rises by $500 million.

Nominal vs Real GDP

- Inflation can distort economic variables like GDP, so we have two versions of GDP:
- **Nominal GDP**
  - values output using current prices
  - not corrected for inflation
- **Real GDP**
  - values output using the prices of a ***base year***
  - is corrected for inflation
  - is used to determine economic growth

Nominal and Real GDP Calculation

	Pizza		Latte	
year	***P***	***Q***	***P***	***Q***
2021	$10	400	$2.00	1000
2022	$11	500	$2.50	1100
2023	$12	600	$3.00	1200

Compute nominal GDP in each year:

Increase:

2021: $10 x 400 + $2 x 1000 = $6,000

2022: $11 x 500 + $2.50 x 1100 = $8,250

2023: $12 x 600 + $3 x 1200 = $10,800

37.5%

30.9%

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Nominal and Real GDP Calculation

	Pizza		Latte	
year	***P***	***Q***	***P***	***Q***
2021	$10	400	$2.00	1000
2022	$11	500	$2.50	1100
2023	$12	600	$3.00	1200

Compute real GDP in each year, using 2011 as the base year:

Increase:

2021: $10 x 400 + $2 x 1000 = $6,000

2022: $10 x 500 + $2 x 1100 = $7,200

2023: $10 x 600 + $2 x 1200 = $8,400

20.0%

16.7%

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Nominal vs. Real GDP

year	*Nominal GDP*	*Real GDP*
2017	$19,543	$18,144
2018	$20,611	$18,688
2019	$21,433	$19,092

In each year,

- nominal GDP is measured using the (then) current prices.
- real GDP is measured using constant prices from a base year.

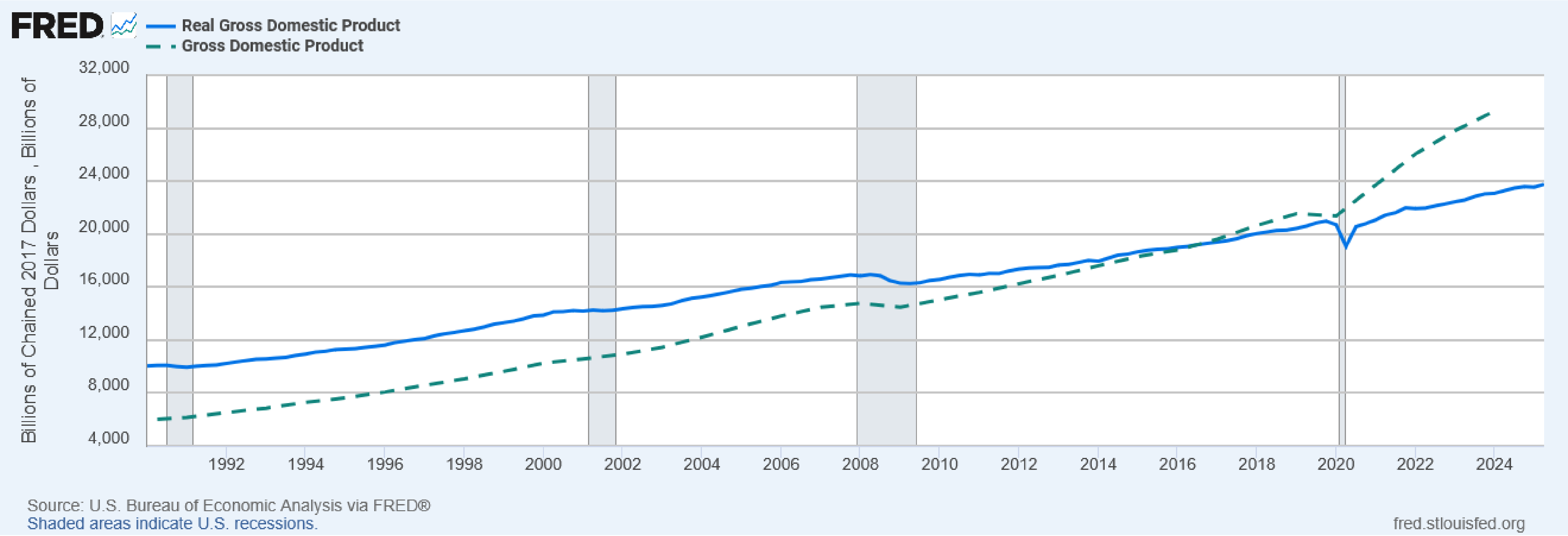
Nominal vs Real GDP Changes

year	*Nominal GDP*		*Real GDP*	
2017	$19,543		$18,144	
2018	$20,611	5.5%	$18,688	2.99%
2019	$21,433	3.99%	$19,092	2.16%

- The change in nominal GDP reflects both prices and quantities.
- The change in real GDP is the amount that GDP would change if prices were constant (i.e., if zero inflation).

Hence, real GDP is corrected for inflation.

Nominal and Real Gross Domestic Product



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The GDP Deflator

- The GDP deflator is a measure of the overall level of prices.
- Definition:

$$\text{GDP deflator} = 100 \times \frac{\text{nominal GDP}}{\text{real GDP}}$$

- One way to measure the economy's **inflation rate** is to compute the percentage increase in the GDP deflator from one year to the next.

The GDP Deflator

- We can also determine nominal GDP and real GDP using the prior definition.

$$\text{Nom GDP} = \text{Real GDP} \times \text{GDP deflator}$$

$$\text{Real GDP} = \frac{\text{Nom GDP}}{\text{GDP deflator}}$$

- Notice the 100 disappeared.
  - That's because now the GDP deflator is out of index form (i.e. 114.6 = 1.146)

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Calculating the GDP Deflator

year	*Nominal GDP*	*Real GDP*	*GDP Deflator*
2017	\$19,543	\$18,144	107.7
2018	\$20,611	\$18,688	110.3
2019	\$21,433	\$19,092	112.3

2.4%

1.8%

Compute the GDP deflator in each year:

2017: 100 x (19543/18144) = 107.7

2018: 100 x (20611/18688) = 110.3

2019: 100 x (21433/19092) = 112.3

Activity: Computing GDP

	2018 (base yr)		2019		2020	
	P	*Q*	*P*	*Q*	*P*	*Q*
Good A	$30	900	$31	1000	$36	1050
Good B	$100	192	$102	200	$100	205

Use the above data to solve these problems:

1. Compute nominal GDP in 2018.
2. Compute real GDP in 2019.
3. Compute the GDP deflator in 2020.

Activity: Computing GDP

	2018 (base yr)		2019		2020	
	P	*Q*	*P*	*Q*	*P*	*Q*
Good A	$30	900	$31	1,000	$36	1050
Good B	$100	192	$102	200	$100	205

1. Compute nominal GDP in 2018. $30 x 900 + $100 x 192 = $46,200
2. Compute real GDP in 2019. $30 x 1000 + $100 x 200 = $50,000
3. Compute the GDP deflator in 2020.
   - [($36 x 1050 + $100 x 205) / ($30 x 1050 + $100 x 205)] x 100= 112.1

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Activity: Calculate Real and Nominal GDP

1. Real GDP is $5,000 and the GDP deflator is 115. What is nominal GDP?
2. Nominal GDP is $6,000 and the GDP deflator is 95. What is real GDP?

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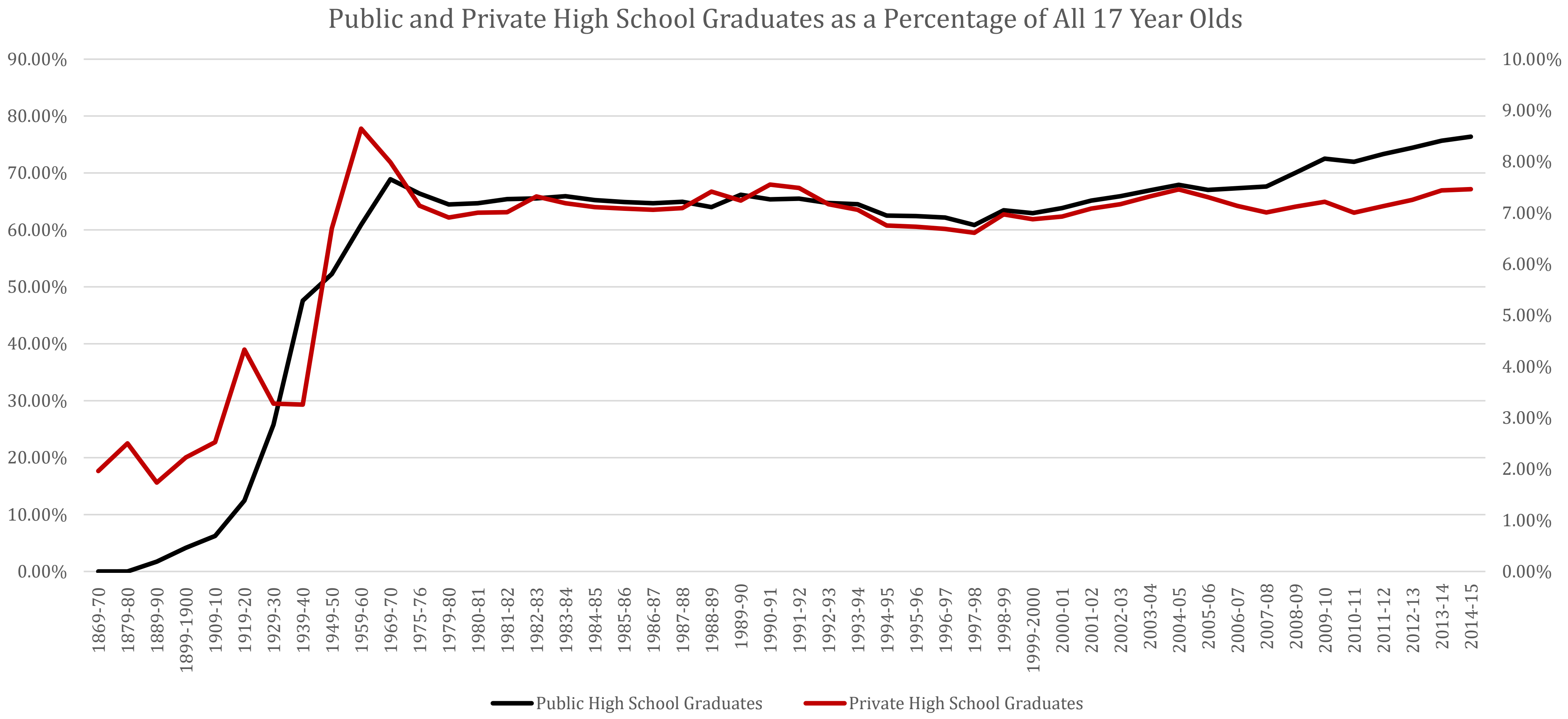
Activity: Calculate Real and Nominal GDP

1. Real GDP x GDP deflator: $5,000 x 1.15 = $5,750
2. Nominal GDP/GDP deflator: $6000 / 0.95 = $6,315.78

GDP and Economic Well-Being

- ***Real GDP per capita is the main indicator of the average person's standard of living.***
- But GDP is not a perfect measure of well-being.
- GDP Does Not Value:
    - the quality of the environment
    - leisure time
    - non-market activity, such as the child care a parent provides his or her child at home
    - an equitable distribution of income

GDP and High School Graduates



Source: NCES Digest of Education Statistics, 2016. Table 219.10

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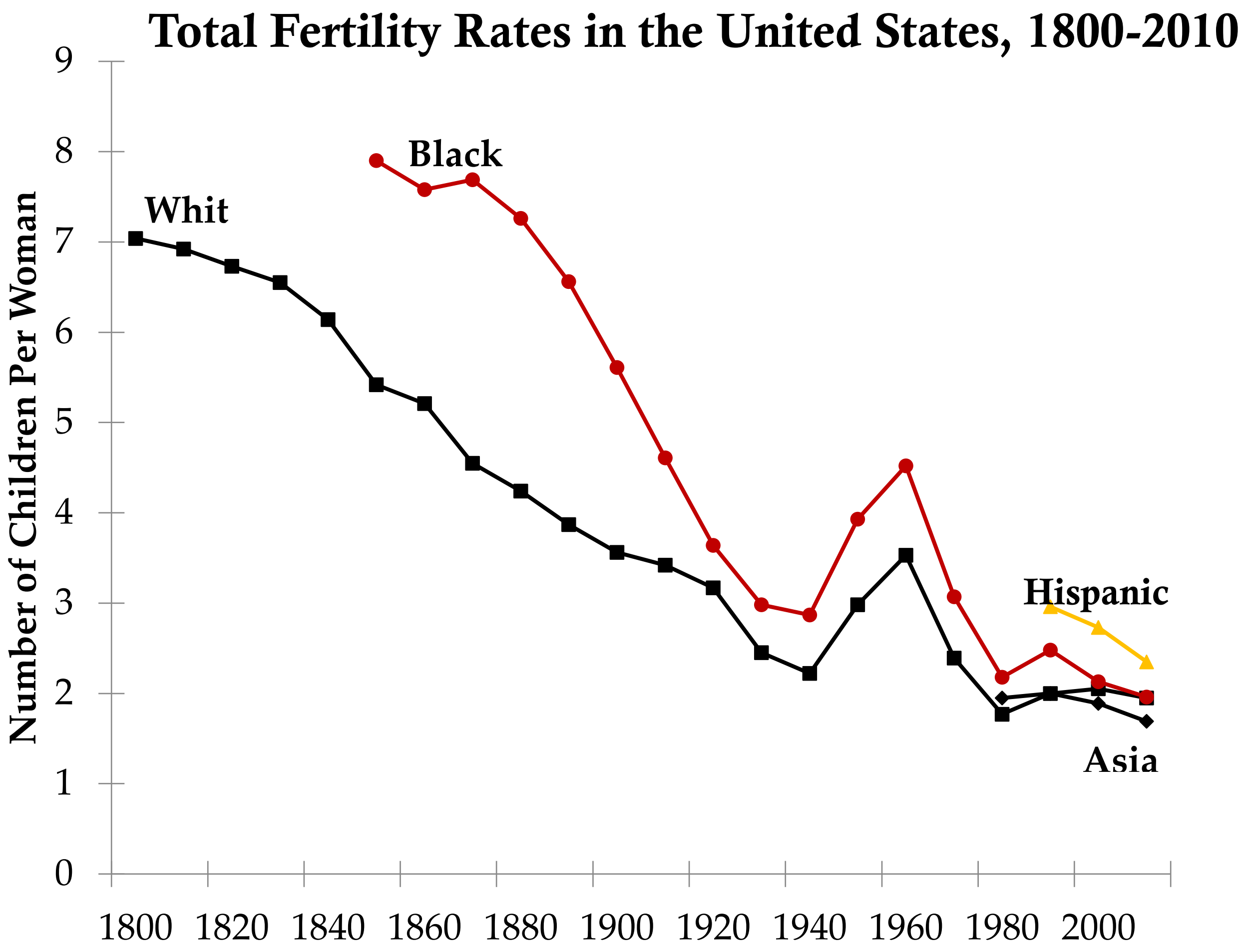
- As GDP increased through time, so did the rate of high school graduations

GDP and Fertility Rates in the US

- As GDP increased over time, the fertility rates of women decreased. Why is that important?



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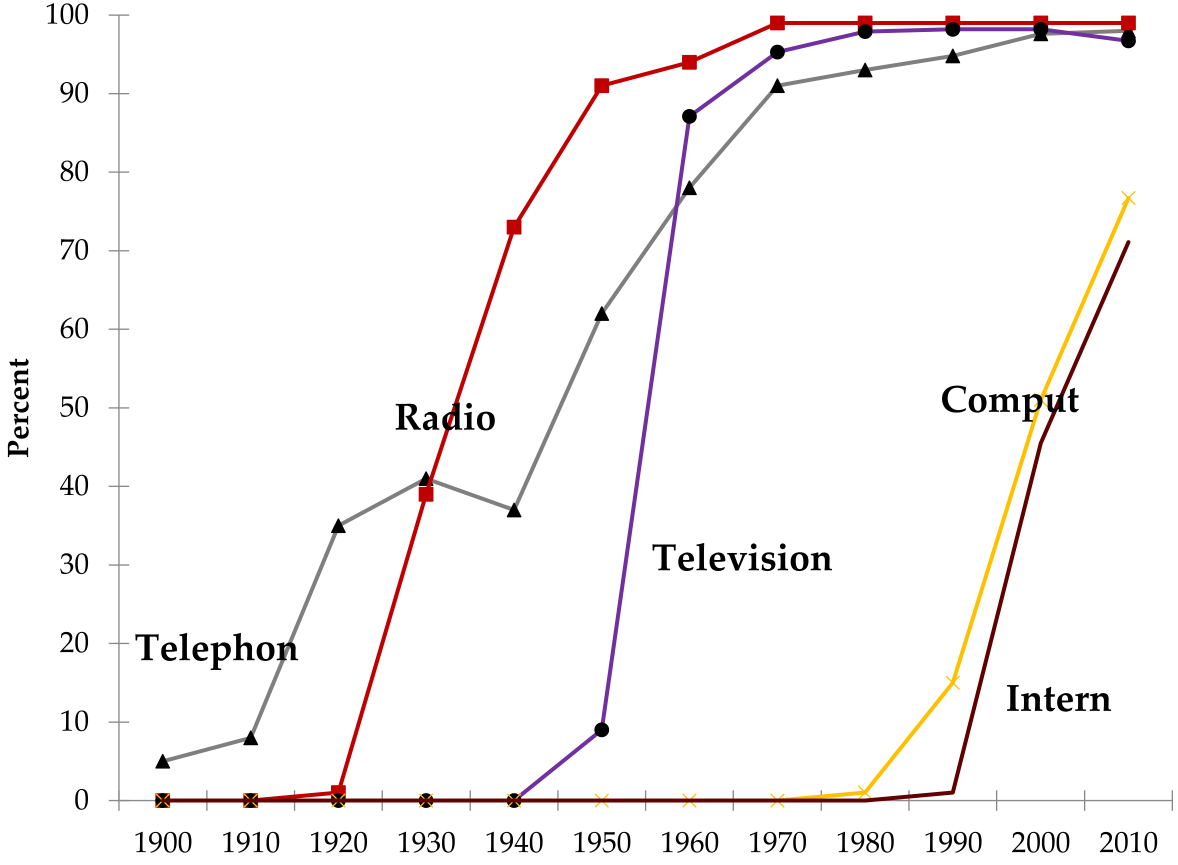
GDP and Communication/Technology

- As GDP increased over time, more communication and technology use increased.

Percentage of U.S. Households with Telephones, Radios, Televisions, Computers, and/or the Internet, 1900–2010



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Then Why Do We Care About GDP?

- Having a large GDP enables a country to afford better schools, a cleaner environment, health care, etc.
- Many indicators of the quality of life are positively correlated with GDP.
- While not a perfect measure of the economy, it does provide a very useful benchmark for economic growth.

Key Takeaways

- GDP measures the market value of all final goods and services produced within a country in a given period.
- It excludes intermediate goods, used goods, and non-market activities.
- GDP can be measured by expenditure, income, or production approaches — and these should match.
- Nominal vs. real GDP separates price changes from actual growth.
- GDP is a powerful but imperfect measure of well-being; it misses inequality, leisure, and environmental quality.

Knowledge Check

1. Are household chores included in GDP?
2. What approach to GDP adds up all spending on final goods and services?
3. What does “real GDP” measure?
4. If nominal GDP rises faster than real GDP, what’s happening?
5. Which is a limitation of GDP as a measure of well-being?

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Knowledge Check Answers

1. No. They are a non-market activity.
2. Expenditure approach
3. Output valued at constant prices, removing inflation effects
4. Prices are rising (inflation).
5. It doesn’t account for non-market activities or income distribution.